The Circumgalactic Medium: Gas Physics to Galaxy Evolution

Yossi Oren Research Statement

1 Motivation

The circumgalactic medium (CGM) is an extended, massive, and thermally-, dynamically- and chemically-complex gas reservoir surrounding galaxies [1]. The bulk of the CGM is understood to be intergalactic inflows shock heated to virial temperatures and thermally supported against collapse, that then cool and accrete onto the galaxy [2, 3]. At the same time, feedback processes from within the galaxy further heat the CGM, regulating gas cooling and accretion, while ejecting interstellar gas out of the galaxy [4–7].

Observations of the CGM, however, are challenging to perform and interpret and therefore do not yet cover all theoretical predictions. Constraints on the thermal state, chemical structure, and mass of the CGM are relatively more available via analysis of ion absorption lines in quasar spectra [e.g. 8–10, and more]. The observed ions can be correlated with either star formation rates [e.g. 11, 12] or the presence of Active Galactic Nuclei (AGN) [e.g. 13, 14] to provide glimpses into interaction between galactic feedback and the CGM. Galactic inflows are quantified through ion absorption lines tracing cooler gas [e.g. 15], some of which showing kinematics that indicate direct inflows onto galaxies [e.g. 16]. More recent improvements in both analysis methods and instrumentation have allowed insight into the CGM via emission surface brightness (SB) profiles from either the hot phase [17] or the cool phase [18, 19] of the CGM, alongside detections of the thermal Sunyaev Zeldovich (tSZ) signal [e.g. 20]. Most recently, analysis of fast radio bursts from select targets also allow studying the mass of their CGM [21].

Additional, powerful tools in our understanding the CGM are numerical methods. Hydrodynamic simulations track billions of dark matter and baryon particles that are allowed to interact and evolve, whose physics is governed through a combination of fluid dynamics and subgrid physics. Cosmological magneto-hydrodynamic simulations like IllustrisTNG [22–26], EAGLE [27, 28], SIMBA [29], or COLIBRE [30, 31] allow the simultaneous evolution of millions of galaxies and provide cosmological context for structure formation. Lastly, a tradeoff between the very computationally expensive simulations and purely analytical models are semi-analytical models (SAMs), in which galaxies are represented by a collection of reservoirs such as the stellar bulge and disk, ISM, CGM, etc., set within "merger trees" that represent halo growth through accretion and merging [e.g. 32, 33]. Simplified models of physical processes track the flows of material between reservoirs, allowing galaxies to evolve. Simulations and SAMs allow the investigation of a wide range of physical processes, both observable and not, providing insights into phenomena for which we do not have a full analytical description.

That being said, our understanding of the CGM is far from complete. The fine details of gas cooling and accretion are still debated, our modeling of feedback processes and their interactions with their environment is not well constrained, and the conclusions we draw from our observations are very sensitive to our assumptions. In the following statement I describe how I contributed to the field during my PhD, and how I plan to continue doing so in the future.

2 PhD Summary

In my PhD I combined post-processing of the IllustrisTNG simulation with physically motivated power law models (PLMs) of the density, temperature, and metallicity of the virialized phase of the CGM as functions of radius. Combining the two, I investigated the thermal state of the CGM and how it is influenced by galactic feedback, quantified the flow of gas into and out of the galaxy through the CGM ("the baryon cycle"), and analyzed the origins and distribution of highly ionized CGM plasma.

In [34] I investigated the thermal pressure of the CGM as sampled by the tSZ signal. By fitting a PLM to stacked tSZ signals of nearby MW-mass galaxies [35] I was able to conclude that the CGM sampled by the stack could maintain hydrostatic equilibrium if its entropy is constant, whereas isothermal conditions are unstable. I have additionally compared the integrated tSZ signal in the TNG simulation with stacked observations from a large halo mass range [20]. The observed normalized tSZ signal begins to decrease as halo masses reach $\sim 10^{13}$ $M_\odot$, whereas the TNG100 signal drops at lower masses and underestimates observations due to the rise of strong kinetic AGN feedback (see Fig. 1).

Together with the galaxy formation group in the Center for Computational Astrophysics at the Simons Foundation, I analyzed the baryon cycle in the TNG100 simulation [36]. I quantified the flow rates of mass, energy, and metals into and out of the ISM and CGM from $z = 10$ up to $z = 0$ and for halo masses spanning $10^{10} - 10^{14.5}$ $M_\odot$, all in the context of the dominant feedback mechanism per mass and redshift. Feedback from supermassive black holes is very energetic, and the onset of AGN jets has strong effects on galaxy growth by preventing inflows, ejecting significant fractions of the CGM baryons, and quenching star formation. My results were later used to calibrate a semi analytic model of galaxy evolution based on the Santa Cruz SAM [37].

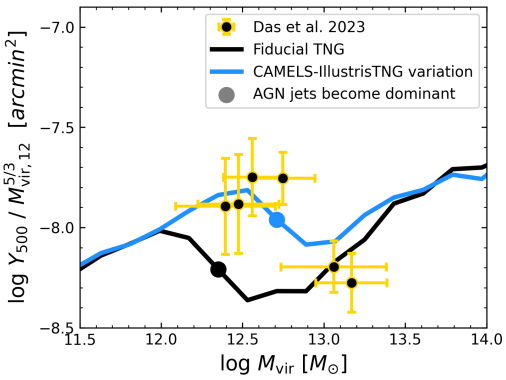


Figure 1: Normalized, integrated tSZ signal. Yellow error bars are the observations by [20], black line is the fiducial TNG100 simulation, and blue line is one of the CAMELS-TNG variations that better match observations. In both simulated profiles, a solid dot marks the onset of AGN jets in the simulation.

I have also investigated the formation and distribution of highly ionized oxygen in the CGM [38]. Using TNG100, I compared the O VI, O VII, and O VIII mean column densities obtained by assuming only collisional ionization equilibrium (CIE) with mean column densities obtained by accounting for both collisional ionization and photoionization by the metagalactic radiation field (MGRF). For each ion, photoionization is significant when the CGM is cooler than the gas temperature where CIE ion production peaks; e.g. O VIII is mainly photoionized in MW-mass halos (see Fig. 2). Comparisons with our PLM of the virialized CGM showed that CIE becomes significant as halos become more massive and their CGM temperature rises. At the same time, growing fractions of the ion mass are collisionally ionized within a less massive, intermediate CGM phase, whose temperature is much lower than the virial temperature but higher than the $\sim 10^4$ K gas accreting onto the galaxy. I was able to quantify the ion masses generated within this intermediate phase from first principles.

I am now in the final stages of a follow up project in which I use the integrated tSZ signal to tighten constrains on implementation of AGN feedback. By investigating variations to subgrid TNG parameters through the CAMELS simulation suite [39–42], I was able to find a group of CAMELS-TNG simulations whose integrated tSZ signals better match observations. Not all CAMELS-TNG simulations in which the integrated tSZ signal approach observations show variations to the same subgrid parameters; yet all relevant simulations delay the onset of kinetic AGN feedback to halo masses closer to 10^{13} $M_\odot$, aligning with predictions from my previous work (see Fig. 1). Most importantly, the galaxy populations in the chosen CAMELS-TNG simulations are still in a good agreement with known relations like the galaxy stellar mass function or stellar to halo mass ratios, adding robustness to my results.

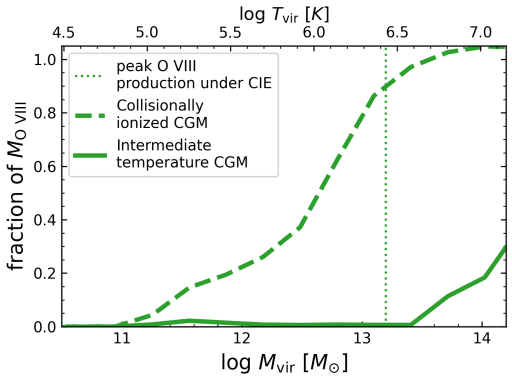


Figure 2: Contributions to the O VIII population in the CGM of TNG100 halos. Dashed line shows the fraction of O VIII that is generated by collisional ionization (in all thermal phases), solid line shows the fraction of O VIII that is generated within the intermediate temperature phase (including CI and PI), and dotted line marks the halo mass where O VIII production under CIE is highest.

Lastly, I am working alongside researchers from the Weizmann Institute of Science on quantifying the bias that may rise in Hubble Parameter (H_0) measurements due to the (mostly ignored) CGM in analyses of gravitational lensing time delays of strongly-lensed quasars. We applied existing lensing pipelines to mock data, using a set of TNG100 massive elliptical galaxies at $z = 0.5$ as lenses with and without accounting for their CGM. We find that an H_0 bias by an order of a few percent arises from the inclusion of a CGM component. The bias is correlated with the mass density of the CGM close to the galaxy, which is in turn correlated with the galaxy star formation rate.

3 Research Plan

In my PhD I have investigated the CGM as a standalone medium and in the context of its relation with both galactic inflows and outflows. The open questions I currently wish to pursue are more focused on processes

related to accretion - CGM cooling via emission and then falling onto the galaxy to fuel star formation.

3.1 Missing Sources of X-ray Emission

The hot, volume filling phase of the CGM emits in UV / X-ray wavelengths through a combination of continuum and line emission. The emissivity of the gas strongly depends on its density, temperature, and metallicity — allowing us to quantify basic properties of the CGM through analysis of SB profiles. As CGM gas cools via emission, luminosity measurements also provide a limit on the accretion rate onto the galaxy. That being said, the luminosity of the CGM is very low due to its diffuse nature, and robust observations of the X-ray emission from the CGM surrounding MW-mass galaxies was only recently obtained by stacking eROSITA observations [17]. Recent numerical models of soft X-ray emission from the CGM surrounding MW-mass galaxies tend to underestimate stacked SB observations, especially at large impact parameters [43]. Additionally, an even more recent interpretation of the X-ray surface brightness measured from the CGM of NGC 4594 underestimates the observed tSZ signal from the same target [44]. The source of the discrepancy is still debated, and may be due to inaccurate numerical models [43], previously unaccounted acceleration of CMB photons by galactic cosmic rays [45], or photoionized gas that is usually neglected [38, 46].

In light of the conclusions from my oxygen ionization paper I wish to explore the latter option, specifically whether the excess X-ray emission around MW-mass galaxies from eROSITA can be attributed to radiative recombination processes. I have carried out timescale calculations to check whether photoionization equilibrium is possible, and found that for e.g. O VII, associated with X-ray lines, photoionization and recombination timescales (in MW-mass halo typical temperatures and densities) are significantly shorter than the Hubble time, in support of this hypothesis. Additionally, initial *CLOUDY* based calculations for MW-mass galaxies in TNG100 suggest that accounting for recombination radiation from photoionized gas results in SB profiles that do not drop off as steeply as CIE based profiles would, with an increase of at least ~ 1 dex in X-ray SB near $R_{\rm vir}$.

3.2 Modes of gas accretion onto galaxies

In the current picture of how MW-mass (and more massive) galaxies form, gas falling from the intergalactic medium onto galaxies is shock heated at the virial radius to virial temperatures, allowing it to remain thermally supported against collapse [2, 3]. If the gas cooling time drops below its free-fall time, it no longer maintains its thermal support and falls onto the galaxy. However, the specifics of this picture are still debated — does the gas cool in bulk, forming a cooling flow onto the galaxy [47]? Do local instabilities lead to cool clouds that precipitate out of the hot phase and onto the galaxy [48, 49]? Or do streams of cool gas simply flow directly through the CGM without being shocked, making their way directly to the galaxy [50]?

I will approach this question by analyzing the energetics of the CGM in cosmological simulations as a function of radius, distinguishing between hot gas that is thermally supported against collapse, cooling gas that is turbulently supported against collapse, and cool gas that is collapsing onto the galaxy — the latter two indicate bulk cooling flows if identified [51]. Conversely, inspection of energy distributions around the mean, alongside high resolution gas density and weighted temperature maps can highlight local cooling instabilities or streams of cold gas penetrating the halo. I have carried out an initial analysis for a small sample TNG50 galaxies, and am looking forward to increase my sample size in order to find

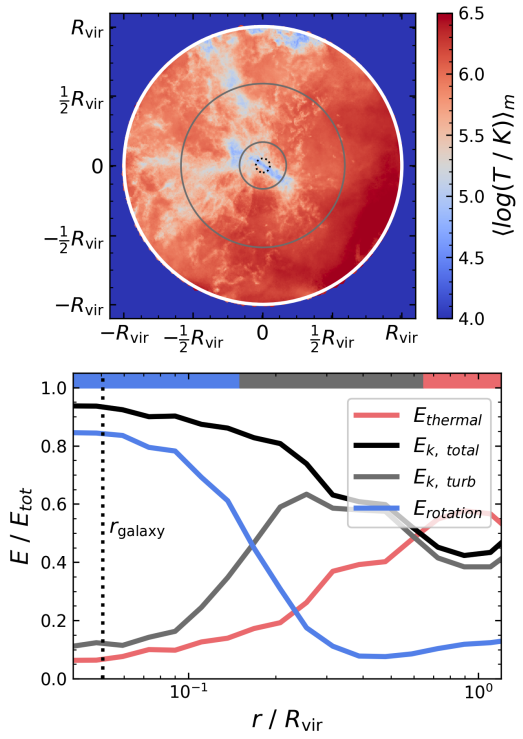


Figure 3: Energy distribution of the CGM surrounding a MW-mass galaxy in TNG50. Top: map of the gas mass weighted log-temperature. Gray circles highlight the region where the energy of the gas is dominated by turbulent motion. Bottom: Individual energy distributions as fraction of the total CGM energy (thermal energy, rotational kinetic energy, turbulent kinetic energy, and total kinetic energy). Upper bar highlights the dominant energy component.

trends and correlations with galaxy and halo populations.

3.3 Tracers of cooling within the CGM

The CGM cooling rate is an important ingredient in determining the accretion rate onto galaxies. I want to infer this cooling rate based on UV absorption observations, informed by simulations.

Our understanding of cooling astrophysical plasmas is that gas cools most effectively when its temperature is $\sim 10^5$ K due to metal line emission [52]. It therefore may be the case that ions whose relative fractions peak at these temperatures — such as O VI — are tracers of gas that is effectively cooling. Under this assumption, the rate by which CGM gas is cooling in MW-mass galaxies can be estimated. By associating the measured O VI column densities with collisionally ionized gas in the virialized phase of the CGM, [53] have come up with an approximation to the energy rates it must lose to cooling. I have found that in TNG100, their estimation of the cooling rates within the CGM of MW-mass galaxies from O VI column densities is consistent with the cooling rates calculated directly for the gas.

However, this is not the case for more massive or less massive galaxies. As I have previously found [38], O VI is neither collsionally ionized nor limited to the virialized phase of the CGM for all halo masses. I will therefore expand the method suggested by [53] to a wider range of galaxy masses by accounting for the thermal phases in which O VI is generated at different halo masses, the mass within each thermal phase, and most importantly the mechanisms by which O VI is ionized in each thermal phase and their associated cooling times.

References

[1] Tumlinson J., Peeples M. S., Werk J. K., 2017, , 55, 389

[2] White S. D. M., Rees M. J., 1978, , 183, 341

[3] Birnboim Y., Dekel A., 2003, , 345, 349

[4] Mathews W. G., Baker J. C., 1971, , 170, 241

[5] Dekel A., Silk J., 1986, , 303, 39

[6] White S. D. M., Frenk C. S., 1991, , 379, 52

[7] Sommer-Larsen J., Gelato S., Vedel H., 1999, , 519, 501

[8] Tumlinson J., et al., 2011, Science, 334, 948

[9] Chen H.-W., et al., 2020, , 497, 498

[10] Tchernyshyov K., et al., 2023, , 949, 41

[11] Borthakur S., et al., 2024, arXiv e-prints, p. arXiv:2409.12554

[12] Flores D., et al., 2026, arXiv e-prints, p. arXiv:2607.19473

[13] Kacprzak G. G., Churchill C. W., Murphy M. T., Cooke J., 2015, , 446, 2861

[14] Berg T. A. M., Ellison S. L., Tumlinson J., Oppenheimer B. D., Horton R., Bordoloi R., Schaye J., 2018, , 478, 3890

[15] Tumlinson J., et al., 2013, , 777, 59

[16] Rubin K. H. R., Prochaska J. X., Koo D. C., Phillips A. C., 2012, , 747, L26

[17] Zhang Y., et al., 2024, , 690, A267

[18] Nielsen N. M., Fisher D. B., Kacprzak G. G., Chisholm J., Martin D. C., Reichardt Chu B., Sandstrom K. M., Rickards Vaught R. J., 2024, Nature Astronomy, 8, 1602

[19] Lin Z., et al., 2025, , 995, 12

[20] Das S., Chiang Y.-K., Mathur S., 2023, , 951, 125

[21] McCarty S., Connor L., Konietzka R. M., 2026, , 1007, L47

[22] Marinacci F., et al., 2018, , 480, 5113

[23] Springel V., et al., 2018, , 475, 676

[24] Nelson D., et al., 2018, , 475, 624

[25] Naiman J. P., et al., 2018, , 477, 1206

[26] Pillepich A., et al., 2018, , 475, 648

[27] Schaye J., et al., 2015, , 446, 521

[28] Crain R. A., et al., 2015, , 450, 1937

[29] Davé R., Anglés-Alcázar D., Narayanan D., Li Q., Rafieferantsoa M. H., Appleby S., 2019, , 486, 2827

[30] Schaye J., et al., 2026, , 548, stag375

[31] Chaikin E., et al., 2026, , 548, stag300

[32] Kauffmann G., White S. D. M., Guiderdoni B., 1993, , 264, 201

[33] Somerville R. S., Primack J. R., 1999, , 310, 1087

[34] Oren Y., Sternberg A., McKee C. F., Faerman Y., Genel S., 2024, , 974, 291

[35] Bregman J. N., Hodges-Kluck E., Qu Z., Pratt C., Li J.-T., Yun Y., 2022, , 928, 14

[36] Oren Y., Pandya V., Somerville R. S., Genel S., Omoruyi O., Sternberg A., 2026, , 999, 259

[37] Omoruyi O., Terrazas B., Oren Y., Gabrielpillai A., Pandya V., Somerville R. S., Sternberg A., Hernquist L., 2026, , 1008, 164

[38] Oren Y., Sternberg A., McKee C. F., Faerman Y., 2026, arXiv e-prints, p. arXiv:2609.15298

[39] Villaescusa-Navarro F., et al., 2021, , 915, 71

[40] Villaescusa-Navarro F., et al., 2023, , 265, 54

[41] Ni Y., et al., 2023, , 959, 136

[42] Genel S., et al., 2026, arXiv e-prints, p. arXiv:2606.10038

[43] Lau E. T., et al., 2025, , 984, 190

[44] Li J.-T., Huang R., Bregman J. N., Wang Q. D., Pratt C., 2026, , 1009, 72

[45] Hopkins P. F., Quataert E., Silich E. M., Sayers J., Ponnada S. B., Sands I. S., 2025, The Open Journal of Astrophysics, 8, 54053

[46] Faerman Y., Sternberg A., McKee C. F., 2020, ApJ, 893, 82

[47] Fabian A. C., 1994, , 32, 277

[48] Sharma P., McCourt M., Parrish I. J., Quataert E., 2012, , 427, 1219

[49] Sharma P., McCourt M., Quataert E., Parrish I. J., 2012, , 420, 3174

[50] Dekel A., et al., 2009, , 457, 451

[51] Stern J., Fielding D., Faucher-Giguère C.-A., Quataert E., 2020, , 492, 6042

[52] Gnat O., Sternberg A., 2007, , 168, 213

[53] McQuinn M., Werk J. K., 2018, ApJ, 852, 33